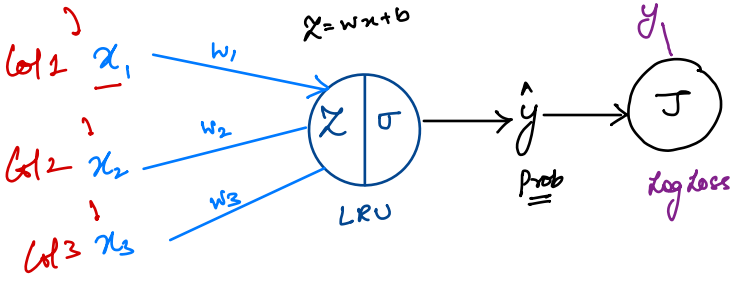
Agenda

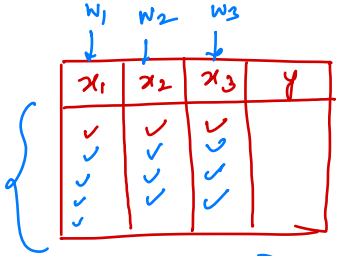
1. Gradients using backpropogation
2. Multi Layer Perceptron
3. What happens .... ? Demonstration.

Training a Neuron



Forward propogation

$$\hat{y} = \sigma\left(w_1x_1 + w_2x_2 + w_3x_3 + b\right)$$



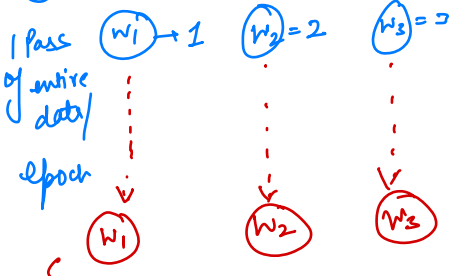
$$z = \vec{w}\vec{x} + b$$

$$a = \sigma(z) = \hat{y}$$

$$= \frac{1}{1+e^{-z}}$$

$a = p = \hat{y}$

1 Pass of entire data / epoch



epoch

Cost function

$$J = -\frac{1}{N}\sum_{i=1}^{N} y \log a$$

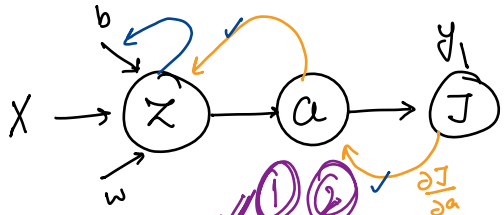
Binary

$$J = -\frac{1}{N}\sum_{i=1}^{N} y \log a + (1-y)\log(1-a)$$

Backpropogation

$$w' = w - \eta \alpha \frac{\partial J}{\partial w}$$

$$b' = b - \alpha \frac{\partial J}{\partial b}$$



$$\frac{\partial J}{\partial w} = \frac{\partial J}{\partial a} \times \frac{\partial a}{\partial z} \times \frac{\partial z}{\partial w} \quad \text{(Chain Rule)}$$

$$\frac{\partial J}{\partial b} = \frac{\partial J}{\partial a} \times \frac{\partial a}{\partial z} \times \frac{\partial z}{\partial b}$$

$$J = -(y \log a + (1-y) \log(1-a))$$

$$\frac{\partial J}{\partial a} = -\left[y \frac{1}{a} + (1-y) \frac{1}{(1-a)} \times (-1) \right]$$

$$\frac{\partial a}{\partial z} = \frac{\partial}{\partial z}\left(\frac{1}{1+e^{-z}}\right) = \frac{\partial}{\partial z}\underbrace{(1+e^{-z})^{-1}}_{x} = \frac{+1 \times +e^{-z}}{\underbrace{(1+e^{-z})^2}_{x}} = \frac{e^{-z}}{(1+e^{-z})} \times \frac{1}{(1+e^{-z})}$$

$$\rightarrow x^{-1} \Rightarrow$$

$$\frac{\partial e^z}{\partial z} \Rightarrow e^z$$

$$\frac{\partial e^{-z}}{\partial z} = e^{-z} \times (-1)$$

$$\frac{\partial x^{-1}}{\partial x} = \frac{1}{x^1} = -x^{-2}$$

$$\frac{e^{-z}+1-1}{(1+e^{-z})} \; a$$

$$\Rightarrow \frac{e^{-z}+1}{1+e^{-z}} - \frac{1}{1+e^{-z}} \; a$$

$$= (1-a)\,a$$

$$\frac{\partial a}{\partial z} = (1-a)a \quad (2)$$

(3)

$$\frac{\partial z}{\partial w} = \frac{\partial (wx+b)}{\partial w} \Rightarrow x$$

$$\boxed{\frac{\partial z}{\partial w} = x}$$

$$\frac{\partial J}{\partial w} = \frac{\partial J}{\partial a} \times \frac{\partial a}{\partial z} \times \frac{\partial z}{\partial w}$$

$$= \underbrace{\frac{-y}{a} + \frac{(1-y)}{1-a}}_{(1)} \times \underbrace{(1-a)a}_{(2)} \times \underbrace{x}_{(3)}$$

$$\Rightarrow \frac{-y(1-a) + a(1-y)}{a(1-a)} \times (1-a)a \times x$$

$$\Rightarrow (-y + ya + a - ya) \times x$$

$$\boxed{\frac{\partial J}{\partial w} \Rightarrow (a-y)x}$$

$$\boxed{\frac{\partial J}{\partial b} \Rightarrow a-y} \times \frac{\partial z}{\partial b} \quad \frac{\partial (wx+b)}{\partial b} \to 0$$

$$\frac{\partial z}{\partial b} = 1$$

$$\frac{\partial J}{\partial W} = \underset{(3m \times 3)}{W} \quad \underset{(3m \times 2)}{X}$$

$$\Rightarrow X^T W$$

$$\Rightarrow (2 \times 3m)\ (3m \times 3)$$

$$\Rightarrow (2 \times 3)$$

$$W^{new} = W^{old} - \alpha \frac{\partial J}{\partial W}$$

$$b = (1,3) \Rightarrow (0,0,0)$$

$$\frac{\partial J}{\partial b} = \frac{\partial J}{\partial a} \times \frac{\partial a}{\partial z} \times \frac{\partial z}{\partial b}$$

$\frac{\partial J}{\partial a} \times \frac{\partial a}{\partial z} = (a - y)$

$\frac{\partial z}{\partial b} = 1$

$(a - y) = dz$

$dz \hookrightarrow (3m, 3)$

In code

$$b^{new} = b^{old} - \frac{\partial J}{\partial b}$$

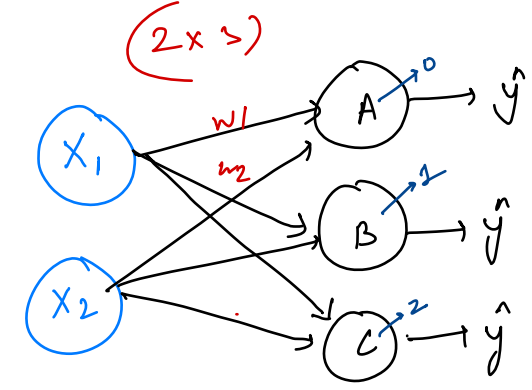
b^{new} : $(1,3)$

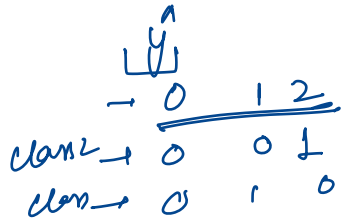
b^{old} : $(1,3)$

$\frac{\partial J}{\partial b}$: $(1,3)$ originally $(3m, 3)$

sum

$(1,3)$

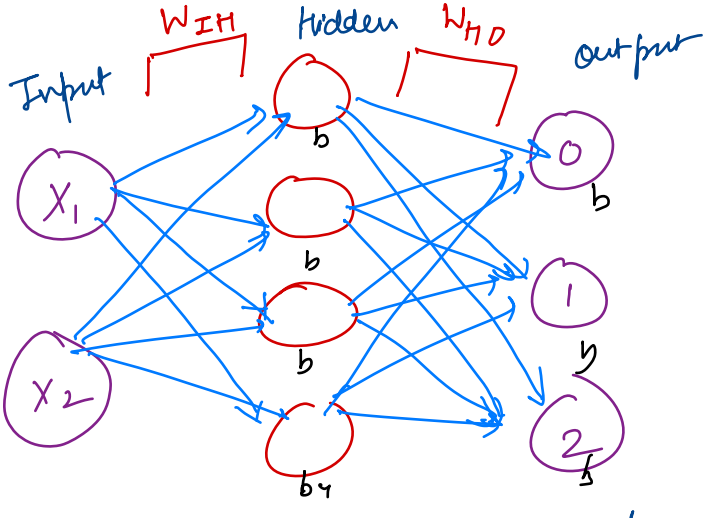




I/p

Output Layer

Without any hidden layer



Dense Layers / Fully Connected Layers

$W_{IH} = 2 \times 4$ $\quad b_H = 1 \times 4$

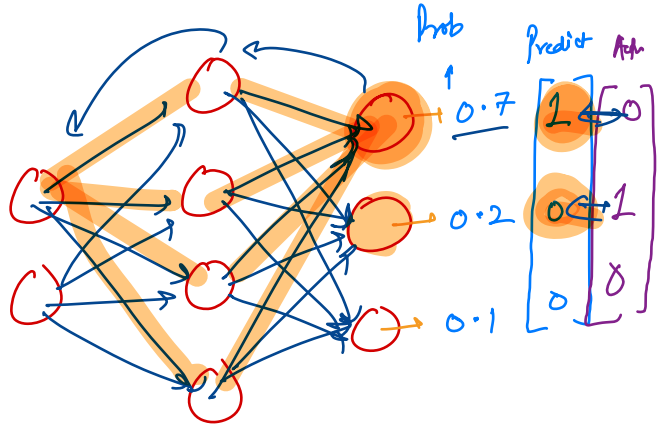
$W_{HO} = 4 \times 3$ $\quad b_O = 1 \times 3$

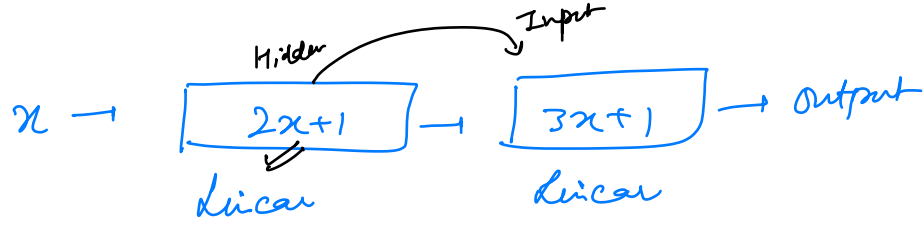
Hidden Layer → Non linearity at least

[Easiest func^n to give non linearity]

→ Computation wise it should be easy

Output → Softmax





$3(2x+1)+1$

$6x+4$